

L04: Camera Parameters

Point from 3D World -> 2D Pixel Coordinate

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = P \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

2D pixel coordinate trans-formation 3D world coordinate

Coordinate Systems

1. World/object coordinate system S_o
 - Written as: $[X, Y, Z]^T$
 - No index -> Object system
2. Camera coordinate system S_k
 - Written as: $[{}^kX, {}^kY, {}^kZ]^T$
3. Image plane coordinate system S_c
 - written as: $[{}^cx, {}^cy]^T$
4. Sensor coordinate system S_s
 - written as: $[{}^sx, {}^sy]^T$

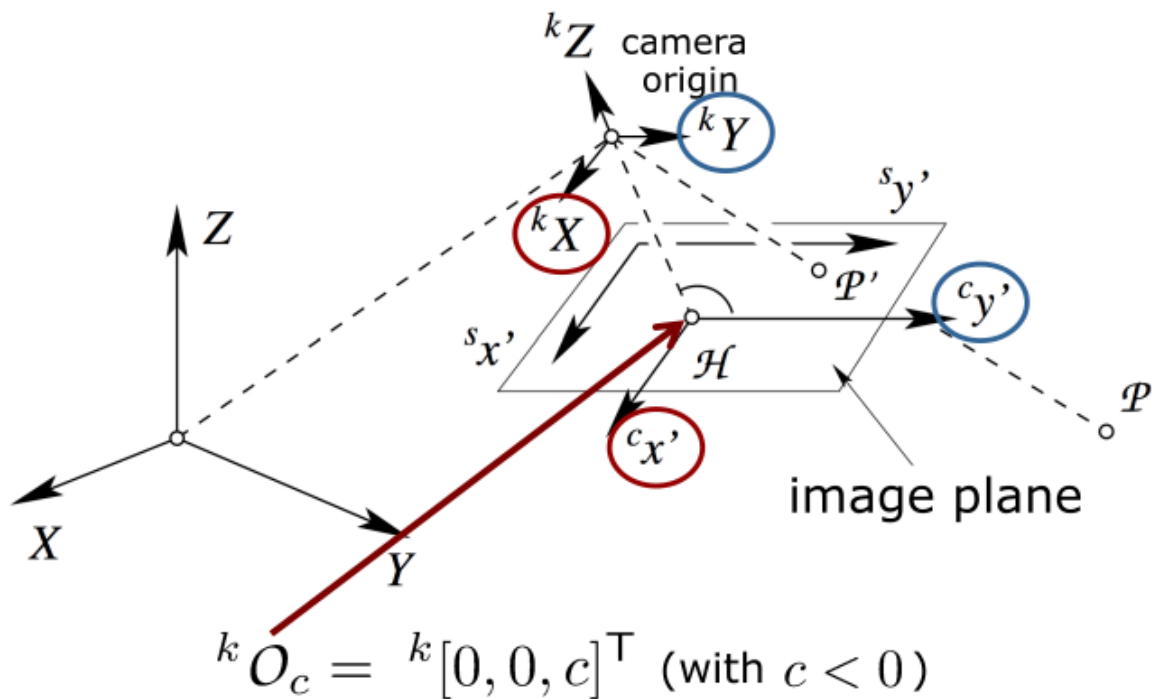
Transformation

$$\begin{bmatrix} {}^s x \\ {}^s y \\ 1 \end{bmatrix} = {}^s H_c {}^c P_k {}^k H_o \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

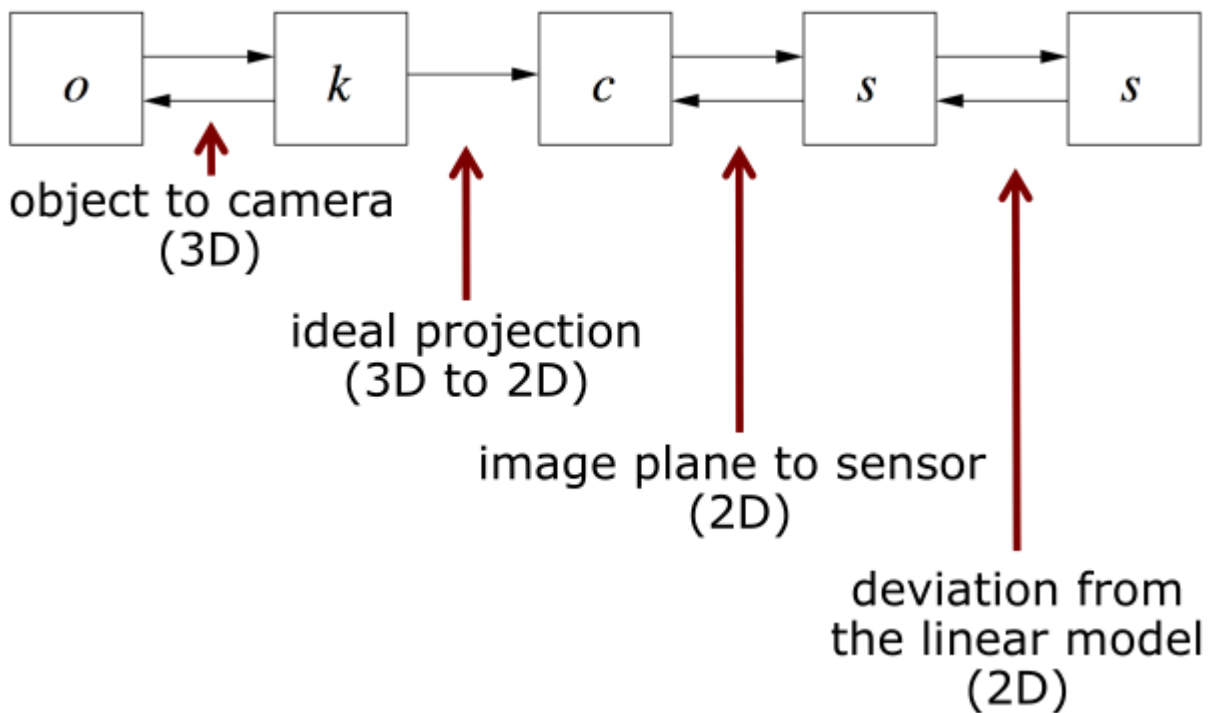
in the sensor system
image plane to sensor
camera to image plane
object to camera
in the object system

Example

The directions of the x- and y-axes in the c.s. k and c are identical. The origin of the c.s. c expressed in k is $(0, 0, c)$

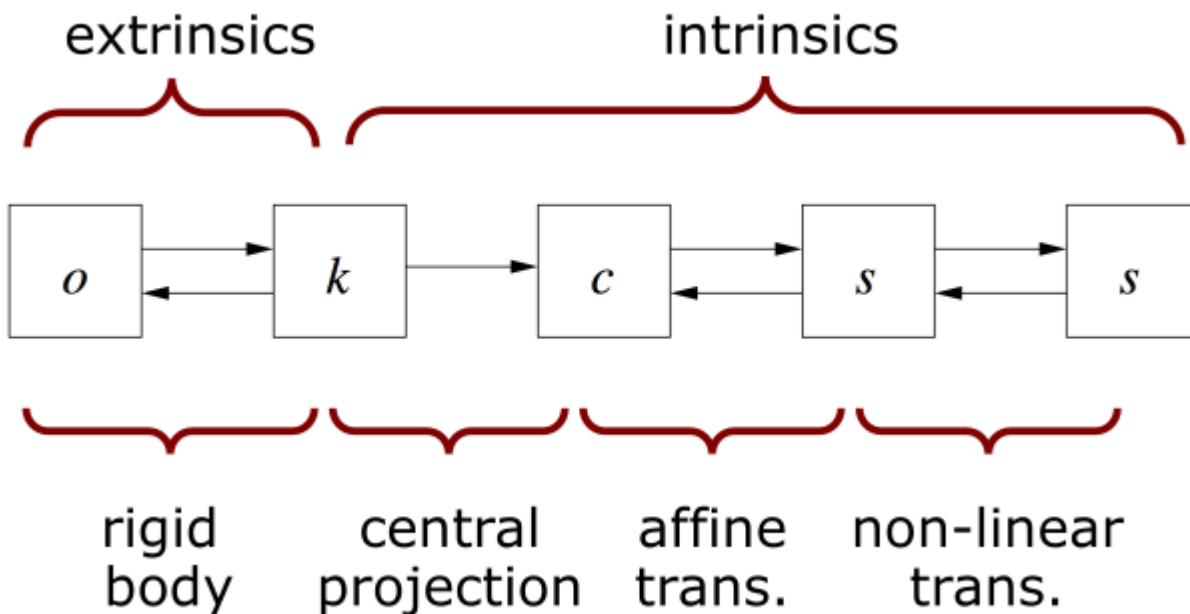


World -> Sensor



Extrinsic & Intrinsic Parameters

- Extrinsic parameters: pose of the camera in the world
- Intrinsic parameters: mapping of the scene in front of the camera to the pixels in the final image (sensor)



Extrinsics

- Pose of camera w.r.t. world
- Invertible transformation
- 6 parameters:
 - 3 position

- 3 heading
- **Translation:** Between origin of the world and camera coordinate systems
- Projection center O of the camera in the world frame
 - $\mathbf{X}_O = [X_O, Y_O, Z_O]^T$
- Rotation R from S_o to S_k
 - ${}^k\mathbf{X}_P = R(\mathbf{X}_P - \mathbf{X}_O)$

In Homogeneous Coordinates

$$\begin{aligned} \begin{bmatrix} {}^k\mathbf{X}_P \\ 1 \end{bmatrix} &= \begin{bmatrix} R & \mathbf{0} \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} I_3 & -\mathbf{X}_O \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{X}_P \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} R & -R\mathbf{X}_O \\ \mathbf{0}^T & 1 \end{bmatrix} \begin{bmatrix} \mathbf{X}_P \\ 1 \end{bmatrix} \end{aligned}$$

Or in Short:

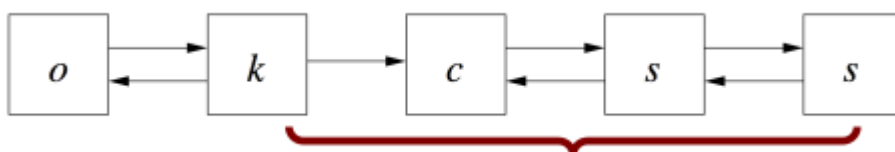
$${}^k\mathbf{X}_P = {}^kH\mathbf{X}_P \quad \text{with } {}^kH = \begin{bmatrix} R & -R\mathbf{X}_O \\ \mathbf{0}^T & 1 \end{bmatrix}$$

Intrinsic Parameters

- Process of projecting points from camera to sensor coordinate system
- **Invertible** transformations:
 - Image plane to sensor
 - model deviations
- NOT invertible:
 - **Central Projection**

Mapping as a 3 Step Process

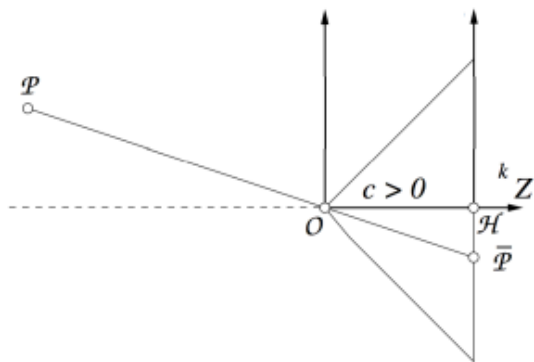
1. **Ideal** perspective projection to the image plane
2. Mapping to the sensor coordinate system
("describes where the pixels are")
3. Compensation for the fact that the 2 previous mappings are idealised



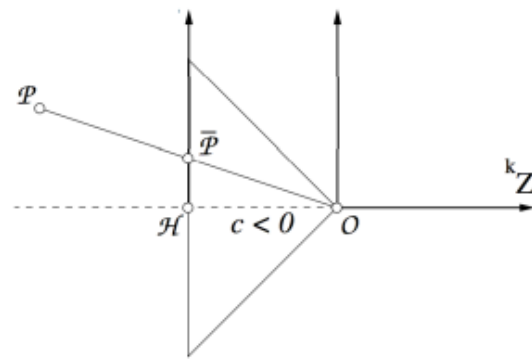
1. Central Projection

Ideal Perspective Projection

- Distortion-free lens
- All rays: straight lines through the projection center $\mathbf{X}_O$
- $\mathbf{X}_O$ is origin of the camera coordinate system S_k
- Focal point O and principal point $\bar{P}$ lie on the optical axis
- Distance from camera origin to image plane $\mathcal{H}$ is constant c

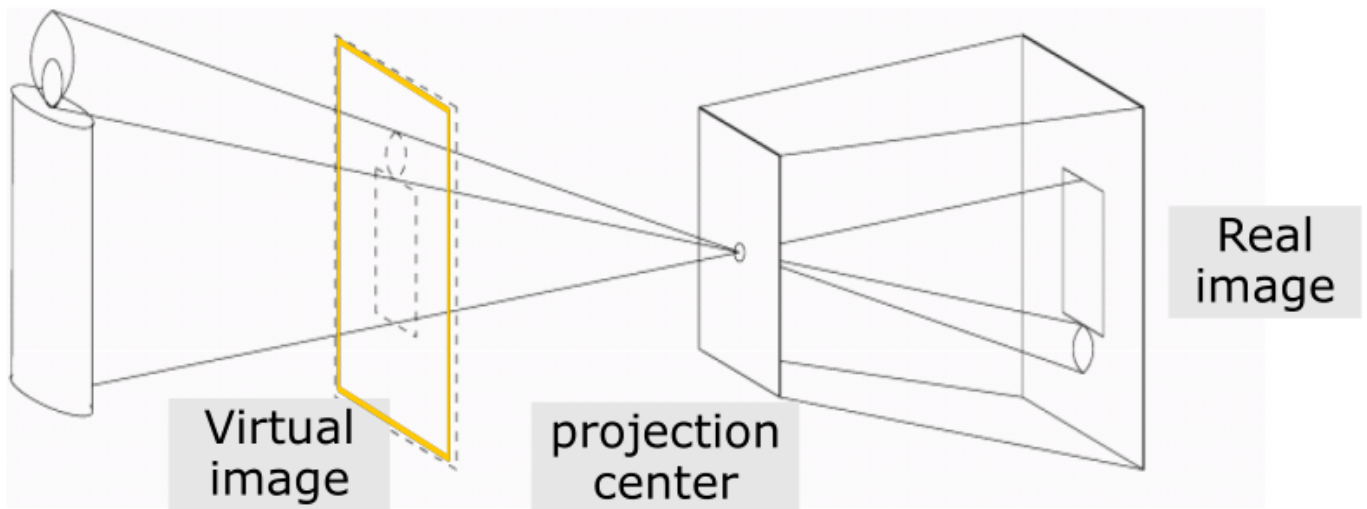


Physically motivated
coordinate system:
 $c > 0$



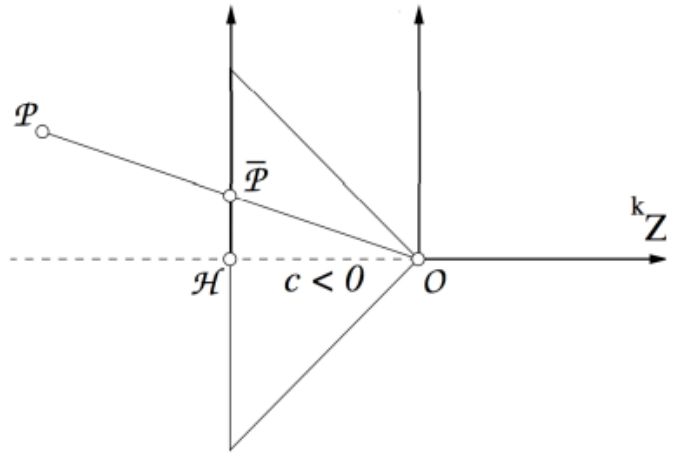
Most popular image
coordinate system:
 $c < 0$

Virtual Image vs Real Image



Ideal Perspective Projection

$$\begin{aligned}
{}^c x_{\bar{\mathcal{P}}} &:= {}^k X_{\bar{\mathcal{P}}} = c \frac{{}^k X_{\mathcal{P}}}{{}^k Z_{\mathcal{P}}} \\
{}^c y_{\bar{\mathcal{P}}} &:= {}^k Y_{\bar{\mathcal{P}}} = c \frac{{}^k Y_{\mathcal{P}}}{{}^k Z_{\mathcal{P}}} \\
\left(c = {}^k Z_{\bar{\mathcal{P}}} = c \frac{{}^k Z_{\mathcal{P}}}{{}^k Z_{\mathcal{P}}} \right)
\end{aligned}$$



In Homogeneous Coordinates

$$\begin{aligned}
{}^c \mathbf{x}_{\bar{\mathcal{P}}} &= \begin{bmatrix} {}^c u_{\bar{\mathcal{P}}} \\ {}^c v_{\bar{\mathcal{P}}} \\ {}^c w_{\bar{\mathcal{P}}} \end{bmatrix} = \begin{bmatrix} c & 0 & 0 & 0 \\ 0 & c & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} {}^k X_{\mathcal{P}} \\ {}^k Y_{\mathcal{P}} \\ {}^k Z_{\mathcal{P}} \\ 1 \end{bmatrix} \\
&= {}^c P_k^k \mathbf{X}_{\mathcal{P}}
\end{aligned}$$

Calibration Matrix

Calibration matrix **for the ideal camera**

$${}^c K = \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Overall Mapping:

- Mapping point to the image plane

$${}^c P = {}^c K R [I_3 | -\mathbf{X}_O] \leftarrow 3 \times 4 \text{ matrix}$$

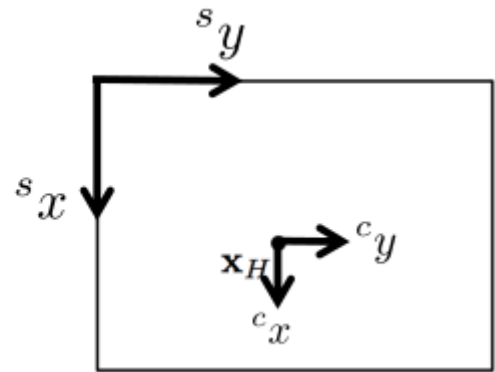
$$\begin{aligned}
{}^c \mathbf{x} &= {}^c P \mathbf{X} \\
&= {}^c K R [I_3 | -\mathbf{X}_O] \mathbf{X} \\
\begin{bmatrix} {}^c u' \\ {}^c v' \\ {}^c w' \end{bmatrix} &= \begin{bmatrix} c & 0 & 0 \\ 0 & c & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & -X_O \\ 0 & 1 & 0 & -Y_O \\ 0 & 0 & 1 & -Z_O \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}
\end{aligned}$$

Linear Errors

- Image to Sensor Mapping
- Location of principal point in image
 - Origin of sensor system is not at principal point

- Compensation through a shift

$${}^sH_c = \begin{bmatrix} 1 & 0 & x_H \\ 0 & 1 & y_H \\ 0 & 0 & 1 \end{bmatrix}$$



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- Shear compensation (s)
 - Typically for digital cameras $s \approx 0$
- Scale difference in x and y based on chip design

$${}^sH_c = \begin{bmatrix} 1 & s & x_H \\ 0 & 1+m & y_H \\ 0 & 0 & 1 \end{bmatrix}$$

Integration into an Affine Transformation

- Calibration matrix as an affine transformation

$$K = {}^sH_c {}^cK = \begin{bmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{bmatrix}$$

- contains 5 parameters:
 - Camera constant c
 - Principal point: x_H, y_H
 - Scale difference: m
 - Shear parameter: s

DLT: Direct Linear Transform

- Mapping $\chi = \mathcal{P}(\chi) : \mathbf{x} = P\mathbf{X}$
- with $P = KR[I_3] - \mathbf{X}_O$

$$\circ \text{ and } K = \begin{bmatrix} c & cs & x_H \\ 0 & c(1+m) & y_H \\ 0 & 0 & 1 \end{bmatrix}$$

- It is model of the **affine camera**
- **Affine camera** = camera with an affine mapping to the sensor c.s. (after applying the central projection)
- Contains **11 parameters**

- 6 extrinsic parameters: $R, \mathbf{X}_O$
- 5 intrinsic parameters: c, x_H, y_H, m, s

Non-Linear Errors

- Real World is **non-linear**
- Reasons:
 - Imperfect lense
 - Planarity of sensor
- Last step covering non-linear effects
- **Location-dependent** shift in sensor coordinate system
- Individual shift for each pixel
- General mapping

$${}^a x = {}^s x + \Delta x({}^s \mathbf{x}, \mathbf{q})$$

$${}^a y = {}^s y + \Delta y({}^s \mathbf{x}, \mathbf{q})$$

- ${}^s \mathbf{x}$ often expressed relative to the principal point (image plane)



Left: not straight line preserving

Right: rectified image

General Mapping in H.C.

$${}^a \mathbf{x} = {}^a H_s({}^s \mathbf{x}) {}^s \mathbf{x}$$

$$\text{with } {}^a H_s({}^s \mathbf{x}) = \begin{bmatrix} 1 & 0 & \Delta x(\mathbf{x}, \mathbf{q}) \\ 0 & 1 & \Delta y(\mathbf{x}, \mathbf{q}) \\ 0 & 0 & 1 \end{bmatrix}$$

→ overall mapping is:

$${}^a\mathbf{x} = {}^aH_s({}^s\mathbf{x})KR[I_3|-\mathbf{X}_O]\mathbf{X}$$

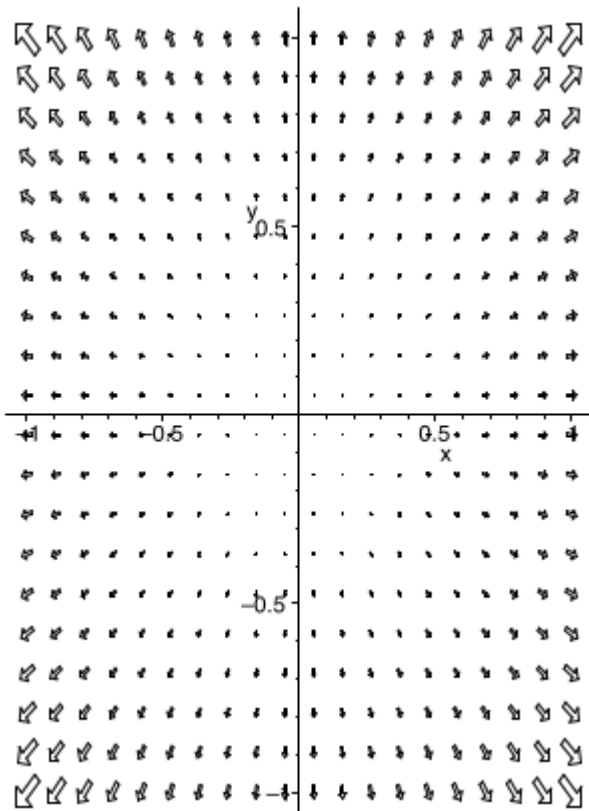
Example Barrel Distortion

- Standard approach for wide angle lenses:
 - Model Barrel distortion

$${}^ax = x(1 + q_1r^2 + q_2r^4)$$

$${}^ay = y(1 + q_1r^2 + q_2r^4)$$

- $[x, y]^T$ point as projected by an ideal pin-hole camera
- r : **distance of pixel** in image to principal point
- q_1, q_2 - additional paramters of the general mapping



Individual Mapping for each point: 2 step Process

1. Projection of the affine camera

$${}^s\mathbf{x} = P\mathbf{X}$$

2. Consideration of non-linear effects

$${}^a\mathbf{x} = {}^aH_s({}^s\mathbf{x}){}^s\mathbf{x}$$

Inverting the Mapping

- Extracting Scene Geometry from the Image
 - Map from image ${}^a\mathbf{x}$ back to the 3D world $\mathbf{X}$

1. ${}^a\mathbf{x} \rightarrow {}^s\mathbf{x}$

- Iterative Solution (unknown $\mathbf{x}$ in ${}^aH_x(\mathbf{x})$)
 - Start with ${}^a\mathbf{x}$ as initial guess
 - iterate $\mathbf{x}^{(\nu+1)} = [{}^aH_s(\mathbf{x}^{(\nu)})]^{-1} {}^a\mathbf{x}$
 - Converges quickly, since ${}^a\mathbf{x}$ is good initial guess

2. ${}^s\mathbf{x} \rightarrow \mathbf{X}$

- Inversion of projective mapping
- Only reconstruction of **ray through** 3D point
- $\mathbf{X} = \mathbf{X}_O + \lambda(KR)^{-1}\mathbf{x}$
- Term $\lambda(KR)^{-1}\mathbf{x}$ describes direction of ray from camera origin $\mathbf{X}_O$ to the 3D point $\mathbf{X}$

Calibrated camera

- **Uncalibrated camera** = intrinsics are **unknown**
- **Calibrated camera** = intrinsics are **known**
- **Metric camera** = intrinsics are **known** and stay **constant**
- **Camera calibration** = process of **obtaining** the intrinsics